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Photoactive polymer coatings for antibacterial applications

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ABSTRACT

Because bacteria can readily induce illnesses in the human body, research into new bactericidal approaches remains relevant to the global community. Photoactive materials (e.g., photocatalysts) provide an environmentally-friendly method for killing bacteria with light. By means of light-mediated reactive oxygen generation, photocatalysts are capable of disrupting bacterial membranes. Much research on antibacterial surfaces has focused on planar substrates or physisorbed coatings, which often provide small active surface areas and limited surface-to-volume ratios, and thereby create a need for new design strategies to incorporate photocatalysts onto more scalable supports. To this end, this contribution describes the synthesis of particle-tethered polymer brush coatings, crosslinked with the organic photocatalyst tetrakis(4-carboxyphenyl)porphyrin (TCPP). Successful engineering on SiO_2 surfaces was confirmed using X-ray photoelectron spectroscopy, tensiometry, and ellipsometry. Singlet oxygen ($^1O_2$) generation was confirmed through dimethyl sulfoxide oxidation experiments under visible (blue) light and bactericidal activity against *Escherichia coli* was verified through live/dead assays and cell viability tests.

1. Introduction

Bacteria create a need for creative scientific solutions to ensure human health. Amongst others, antibacterial materials find widespread use in medical applications, [1,2] water remediation, [3,4] and waste management [5].

Decontaminating potable drinking water is a critical challenge globally [6]. Current large-scale purification processes often rely on chemical additives or separation processes using microfiltration (MF) and ultrafiltration (UF) membranes. Adding disinfecting chemicals imparts detrimental environmental implications and membrane performance is compromised over time due to fouling, which requires cleaning either through chemical additives or backwash. As an alternative, photocatalytic processes can be leveraged to inactivate and kill bacteria [7]. Industrial wastewater processes currently already utilize metal oxides (e.g. ZnO, TiO_2) as photosensitizers to degrade contaminants and inactivate bacteria [8,9]. However, such metal oxide photocatalysts are limited to absorption of UV light (only 3–5 % of the solar spectrum), [10] and introduce challenges related to scalability (limited light-penetration). This limitation has spawned interest in approaches using visible light photocatalysts for anti-bacterial treatment of water [11]. Transition metal-based complexes and fully organic photocatalysts can both absorb visible light, while the latter adds the benefit of not requiring expensive or toxic metals to be effective [12,13]. This can decrease cost and avoid leaching of toxic metals into the environment. Furthermore, heterogeneous approaches exist that immobilize the photocatalysts, [14] e.g., within frameworks, [15] on nano-[16] or micron-scale particles, [17–19] or confined within a polymeric material [20].

Beyond waste-water remediation, photosensitizers find wide-spread use in health care. Light in general is widely used for imaging,[21,22] ultraviolet decontamination, [23] and photothermal therapy [24]. In photodynamic therapy (PDT), photosensitizer molecules are used for photo-generation of reactive oxygen species – e.g., singlet oxygen ($^1O_2$) – which can introduce bactericidal properties by disrupting bacteria cell membranes indiscriminately [25–27]. Various organic dyes (e.g., porphyrins, [28] chlorins, [29] xanthene dyes, [30,31] phthalocyanines, [32] etc.) are capable of generating $^1O_2$ through exciting electrons via

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photon absorption and passing the energy to $^3O_2$. Tetrakis(4-carboxyphenyl)porphyrin has captured attention for its low mammalian cell toxicity, visible light absorption, and facile coordination with metal ions, leading to uses in cancer treatment [33–35].

It becomes evident that new approaches towards antibacterial materials remain as relevant as ever – especially considering that antibiotic resistance in humans is on the rise [3,36]. The need to keep surfaces sterile and decontaminate waste becomes increasingly critical for the well-being of society [3,5]. To this end, anti-fouling, [37] antimicrobial, [38] and anti-bacterial [39] properties can be accomplished both through physisorption [40] (e.g., commercial anti-bacterial paints) and chemisorption [41–43]. Chemisorption provides benefits with respect to longevity of the coatings; the covalent tether between the organic or polymeric coatings and surface can reduce delamination and wear. For polymer-based coatings, chemisorption approaches include grafting-to [44] and grafting-from, [45] e.g., via surface-initiated (SI) polymerization [46]. In recent years, the potential impact for SI polymerization has increased – in part through developments in SI reversible deactivation radical polymerizations (SI-RDRPs) [46–50]. SI-RDRP has become increasingly user-friendly through oxygen-tolerant chemistries [50] and the introduction of external regulation [51] – now affording functionalization of large surface areas, [52] access to a plethora of chemical species, [49,53] the ability to provide spatiotemporal patterning, [54–56] or coatings that respond to external stimuli [57,58].

SI-RDRP now provides access to surface-tethered anti-fouling [59,60] and anti-bacterial surfaces [48]. SI-RDRP offers tunability over surface roughness, coating thickness, functionalization, and more when compared to free radical polymerizations [47,61]. Neutral polymer brushes like poly[ethylene glycol] [62] and poly[2-hydroxyethyl methacrylate] [63] work well as anti-fouling layers on surfaces due to their hydrophilicity but are not bactericidal. Cationic [64,65] and zwitterionic [66] polymer brushes offer bactericidal activity through electrostatic interactions with bacterial membranes. In recent examples, Sroka et al. grafted poly[2-(dimethylamino)ethyl methacrylate] brushes on glass in aqueous solutions of honey via surface initiated atom transfer radical polymerization (SI-ATRP) [61]. Subsequent quaternization of tertiary amino groups to form cationic polymer brushes yielded polymer brushes that were more than 98 % effective against *S. aureus* and *E. coli* [61]. However, intrinsically bactericidal polymer brushes (e.g., cationic or zwitterionic polymer brushes) can also show toxicity towards mammalian cells [48,64]. Coatings responsive to external stimuli (e.g., pH, [67] temperature, [68] light, [69] etc.) can give control over bactericidal activity. Choi et al. grafted poly[di(ethyleneglycol) methyl ether methacrylate] via SI-ATRP which showed tunable thermos-responsiveness based on coating thickness [68]. The temperature response allowed for a triggered release of antibiotic loaded in the polymer brushes. Our group recently used SI-RDRP to engineer photoactive anti-bacterial coatings [69]. Poly[2,2,2-trifluroethyl acrylate] brushes were grafted via surface initiated photoinduced electron transfer reversible addition-fragmentation chain transfer (SI-PET-RAFT), during which zinc tetraphenylporphyrin (ZnTPP) was incorporated into the polymer backbone while catalyzing the polymerization [69]. Control studies showed that the polymer brushes with ZnTPP incorporated were only bactericidal when irradiated with green light [69].

While these studies are promising scientific advances, much work in this field remains focused on flat substrates. For applications such as water-remediation or PDT, however, non-planar substrates are preferred to increase surface-to-volume ratios and improve scalability. Furthermore, the silane-anchor in polymer brushes is prone to hydrolysis – even in humid air, [19,70] introducing concerns about contamination and longevity of polymer brush-based antibacterial approaches.

To address these challenges, this contribution describes a crosslinked photoactive polymer brush approach for antibacterial applications (see Fig. 1). The crosslinking methodology forms a photoactive shell around solid glass supports to increase surface-area and prevent leaching of surface-incorporated photocatalysts. These functionalized materials were tested for photogeneration of reactive $^1O_2$ and related non-specific antibacterial activity.

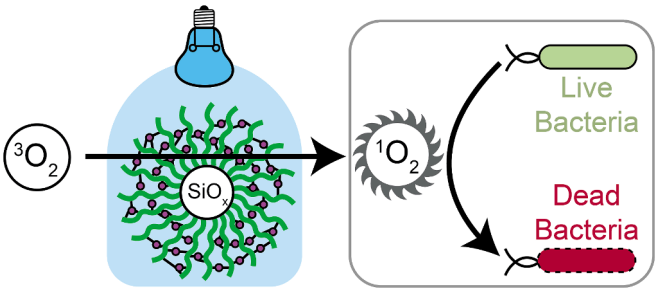

Fig. 1. Scheme of photocatalyst functionalized glass beads catalyzing singlet oxygen and destroying bacteria.

2. Results and discussion

2.1. Bead synthesis

Crosslinked antibacterial polymer brush-based materials were engineered via surface-initiated polymerization. To this end, silica beads (SiO_2; $D_z = 76.3$ μm) were modified with a chain transfer agent (CTA) for surface initiated reversible addition-fragmentation chain transfer (SI-RAFT) polymerization. The CTA 4-cyano-4-[(dodecylsulfanylthiocarbonyl)sulfanyl]pentanoic acid (CDTPA) was immobilized following previously reported procedures [71]. From this CDTPA initiating layer, SI-RAFT was performed to synthesize poly(2-hydroxethylacrylate) (PHEA) brushes. PHEA was chosen for its crosslinking capabilities with tetrakis(4-carboxyphenyl)porphyrin (TCPP) through esterification.

The surface-tethered PHEA brushes (PHEA@SiO_2) were characterized before and after TCPP-crosslinking (see Fig. 2**a**) by X-ray photoelectron spectroscopy (XPS), spectroscopic ellipsometry, and tensiometry. Fig. 2**b** shows the XPS spectrum of PHEA brushes, indicating oxygen (O1s; $BE_{O1s} = 534.3$ eV), carbon (C1s; $BE_{C1s} = 285$ eV), and silicon peaks (Si2s; $BE_{Si2s} = 155.2$ eV and Si2p; $BE_{Si2p} = 103.3$ eV). The experimental C1s:O1s peak ratio of 1.8:1, closely matched the theoretical ratio of 1.7:1 within fitting errors. For high-resolution C1s photoelectron spectra (Fig. 2**c**), distinct carbon binding environments are expected for PHEA brushes at 2:2:1 for C–C:C-O:C = O. In good agreement, the experimental ratios were determined as 1.9:2:1 for C–C:C-O:C = O with peaks at 285.0 eV, 286.5 eV, and 288.6 eV, respectively. Following established SI-RAFT protocols, [71] polymerization of PHEA occurs both in solution and from the surfaces. This allows characterization of the solution PHEA polymer via ^{1}H NMR spectroscopy to give $M_n = 15{,}000$ g mol^{-1} (see **Figure S1**). To facilitate characterization of PHEA brush thicknesses, the same experimental conditions were used to modify planar SiO_2 wafers (2x2 cm^2). The PHEA brush thickness was measured at $d = 15.2 \pm 0.5$ nm by ellipsometry. Water contact angles (WCA) were measured as $\theta_{SiO2} = 25 \pm 2^\circ$ and $\theta_{PHEA@SiO2} = 54 \pm 2^\circ$ before and after PHEA brush modification, respectively, further indicating successful growth of the PHEA brushes (see **Figure S2**).

To confer antibacterial properties to PHEA@SiO_2, tetrakis(4-carboxyphenyl)porphyrin (TCPP) was used to crosslink the polymer brushes. TCPP was synthesized from 4-carboxybenzaldehyde and pyrrole in propionic acid, which yielded a purple solid. ^{1}H NMR spectra for the final product confirm successful synthesis with the appearance of a signature peak of TCPP at δ = –2.93 ppm (s, 2H, NH; see **Figure S3**). The four carboxylic acid groups on TCPP were used to crosslink the PHEA brushes via Steglich esterification. In detail, PHEA@SiO_2 beads were immersed in a TCPP solution with *N*,*N*'-dicyclohexylcarbodiimide (DCC) and 4-dimethylaminopyridine (DMAP). After 48 h, the beads were rinsed, yielding TCPP-crosslinked PHEA@SiO_2 brushes (X-TCPP@SiO_2).

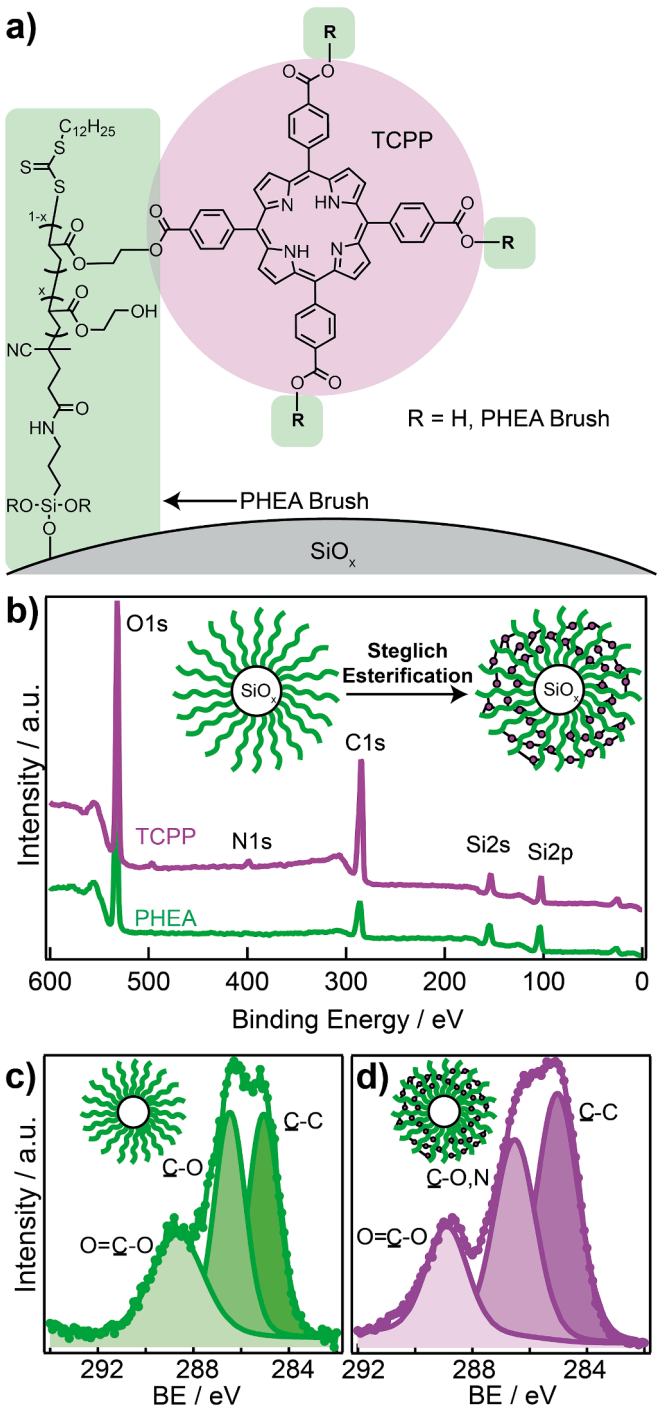

Fig. 2. (a) Poly(hydroxy ethyl acrylate) (PHEA) polymer brushes on SiO_2 beads and crosslinked using tetrakis(4-carboxyphenyl)porphyrin (TCPP). (b) X-ray photoelectron spectra of PHEA brushes before (bottom, green) and after TCPP-crosslinking (top, purple). High-resolution carbon C1s spectra of (c) PHEA polymer brushes and (d) TCPP-crosslinked PHEA polymer brushes.

Beyond the now faint purple color (see Figure S4), the presence of TCPP on X-TCPP@SiO_2 was verified via XPS. A nitrogen N1s peak appeared at $BE_{N1s} = 400$ eV (see Fig. 2**b**) which can be attributed to the surface-tethered porphyrin (TCPP, see Fig. 2**a**). The atomic peak ratios were measured as 32.4:11.6:1 for C1s:O1s:N1s respectively. This structure also contributes to an elevated C–C peak in the carbon high resolution spectra (see Fig. 2**d**). The peak ratio in the high resolution spectra is found to be 2.3:1.8:1 for C–C:C-(O,N):C = O. Via high-resolution C1s peak fitting and taking into account the expected peak ratios for PHEA (2:2:1) and TCPP (9:2:1), the TCPP incorporation in X-TCPP@SiO_2 was calculated as 4 %. UV–vis studies provided another estimate of the incorporation of TCPP on X-TCPP@SiO_2 by studying the crosslinking solution. 86 ± 6 % of feed TCPP (by mass) were determined in the leftover crosslinking solution after treating PHEA@SiO_2 (see Figure S5). This suggests a maximum TCPP incorporation of 20 % by mass into X-TCPP@SiO_2. However, considering our estimate from XPS, this likely includes physiosorbed photocatalyst or other means of TCPP mass loss throughout the reaction. UV–vis analysis suggests that 250 μg TCPP is incorporated per 1 g of X-TCPP@SiO_2. Water stability studies on X-TCPP@SiO_2 showed minimal leaching of TCPP after seven days (see Figure S6). Again, analogous chemistries were performed on planar SiO_2 substrates to permit characterization via ellipsometry and tensiometry. As per our expectations, the PHEA brush thickness increases after TCPP-crosslinking from $d_{PHEA} = 15.2 \pm 0.5$ nm to $d_{X\text{-}TCPP} = 18.3 \pm 0.9$ nm. The water contact angle (WCA) also increases from $\theta_{PHEA@SiO2} = 54 \pm 2^\circ$ to $\theta_{X\text{-}TCPP@SiO2} = 69 \pm 2^\circ$, indicating an expected increase in the amount of hydrophobic carbonaceous material within the film (see Figure S2).

2.2. Singlet oxygen tests

Because singlet oxygen ($^1O_2$) is known to be bactericidal by non-specifically interrupting cell membranes, [25–27] X-TCPP@SiO_2 beads were interrogated regarding their capability of generating $^1O_2$. Dimethyl sulfoxide (DMSO) is a $^1O_2$ scavenger through conversion to dimethyl sulfone ($DMSO_2$). The reaction is observable through ^{1}H NMR with a shift from δ = 2.5 ppm (s, 6H, CH_3; see Figure S7) for DMSO to δ = 3.0 ppm (s, 6H, CH_3; see Figure S7). As such, the rate of $DMSO_2$ production can provide insight into $^1O_2$ generation. Fig. 3 shows that X-TCPP@SiO_2 beads generated $DMSO_2$ at a rate 550 ± 20 μg h^{-1} under blue light irradiation (λ_{max} = 470 nm, 2 mW cm^{-2}). TCPP in solution shows a lower production rate of 440 ± 40 μg h^{-1} at a photocatalyst concentration of 0.25 mg mL^{-1}. This TCPP concentration was studied as an equivalent reaction to approximately match the photocatalytic conditions for X-TCPP@SiO_2. We found the performance of X-TCPP@SiO_2 exceeding TCPP in solution for $^1O_2$ generation curious. Indeed, previous work from our group and others has shown benefits of immobilizing photocatalysts [16,18]. To date, we cannot give unambiguous scientific evidence at this point, but possible influencing factors include the chemical environment surrounding the TCPP (polymer brushes) [72,73] as well as differences in photobleaching behavior between solutions and immobilized catalysts [74]. As a control experiment, pure DMSO (without added catalysts) oxidized at a rate of 48 ± 3 μg h^{-1}, and in the presence of unfunctionalized beads, this rate decreased to 28 ± 2 μg h^{-1}. Based on these studies, the X-TCPP@SiO_2 beads showed improved capabilities of photo-generating $^1O_2$ when compared to non-functionalized controls.

2.3. Antibacterial tests

To test the bactericidal capabilities of the crosslinked X-TCPP@SiO_2 beads, *E. coli* (strain BW25113) cells suspended in phosphate buffered saline (PBS) solution were treated with X-TCPP@SiO_2 beads for 24 h under blue light (λ_{max} = 470 nm, 2000 μW cm^{-2}). After treatment, live/dead staining assays (see Fig. 4) and cell viability tests (see Fig. 5) were performed to assess the effects on the bacteria cells. Control experiments were designed to examine whether the observed bactericidal activity was due to the $^1O_2$ produced by irradiation of the X-TCPP@SiO_2 beads. Thus, bacteria were also treated by irradiation only ("no bead" condition) and in the presence of X-TCPP@SiO_2 beads but in absence of light. These experiments aimed to interrogate light exposure and toxicity from TCPP exposure.

Live/dead assays employ a green-fluorescent nucleic acid stain (SYTO 9), which labels all bacteria, and a red-fluorescent nucleic acid stain (propidium iodide), which can only penetrate and label bacteria

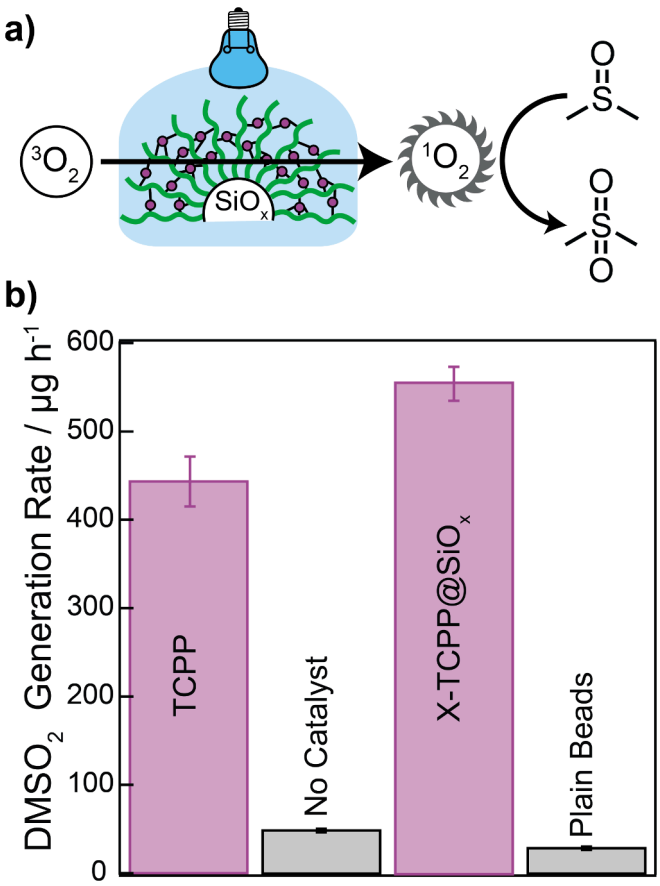

Fig. 3. (a) Scheme of dimethyl sulfoxide (DMSO) to dimethyl sulfone ($DMSO_2$) via photo-oxidation using X-TCPP@SiO_2 and its ability to generate singlet oxygen ($^1O_2$). (b) $DMSO_2$ production rates (after 24 h) for TCPP in solution, absence of any photocatalyst (pure DMSO), X-TCPP@SiO_2 polymer brushes, as well as unfunctionalized SiO_2 beads.

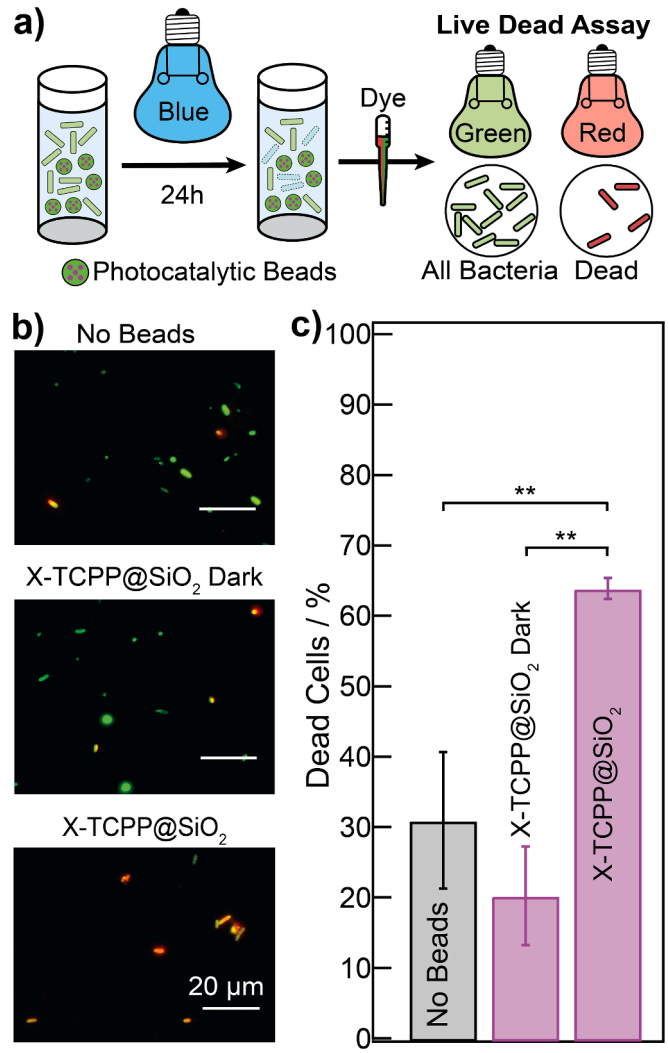

Fig. 4. (a) Scheme of live/dead assay conducted on *E. coli* BW25113 cells exposed to photocatalytic SiO_2 beads and blue light. (b) Fluorescent microscopy images from no beads, X-TCPP@SiO_2 beads with no light, and X-TCPP@SiO_2 beads tests (top to bottom). Green and red indicate cells staining positive for SYTO 9 and propidium iodide respectively. (c) Percentages of bacteria staining positive for propidium iodide indicating cell death, $^{**}p < 0.01$.

with damaged membranes (see Fig. 4**a**). Cells that stained red in fluorescence microscopy were scored as dead (see Fig. 4**b**). The number of dead cells and the total number of cells were counted to determine the percentage of cells staining positive for propidium iodide in each treatment condition (see Fig. 4**c**). Treatment with irradiated X-TCPP@SiO_2 beads induced death in 64 ± 1 % of cells. This was statistically higher than the percent death induced by all other control conditions. When the cells were not exposed to the beads, only 30 ± 10 % of cells died. When exposed to X-TCPP@SiO_2 in the dark, 21 ± 7 % of cells died.

In some instances, bacteria labeled as "live" in this assay may not be able to proliferate. Therefore, viability assays were performed to confirm that the TCPP treatment is not only increasing apparent cell death, but also attenuating bacteria proliferation. To monitor viability, cells were diluted to enable colony counting and grown on agar plates for 24 h post-treatment (see Fig. 5**a**). Treatment of *E. coli* bacteria with X-TCPP@SiO_2 beads in the presence of blue light results in no colony forming units after 24 h, which is significantly lower than all other control conditions. In the "no beads" and X-TCPP@SiO_2 beads without light conditions, the live/dead assay shows low cell death ("no beads" = 30 ± 10 % cell death; X-TCPP@SiO_2 beads without light = 21 ± 7 % cell death) and the viability assay shows high viability ("no beads" = (1.56 ± 0.99) x 10^7 CFU mL^{-1} and X-TCPP@SiO_2 beads without light = (1.2 ± 0.9) x 10^7 CFU mL^{-1}), suggesting that the observed bactericidal activity in the TCPP treated cells is not due to light exposure or TCPP toxicity (see Fig. 5**b, c**). Together, the live/dead assay and viability tests suggest that treatment with X-TCPP@SiO_2 beads in the presence of blue light promotes cell death and renders the bacteria unable to proliferate.

Viability was also compared to exponential cell growth, which represents the number of cells present in growth media following no manipulation. The differences between cell count in exponential growth compared to "no beads" and X-TCPP@SiO_2 beads without light are small (see Fig. 5**c**), suggesting that suspension in PBS, gentle stirring, and centrifugation do not significantly impact cell viability. Physical damage from bead collision, intrinsic toxicity of the photocatalytic coating, and damage from blue light irradiation can all be dismissed as significant causes of bactericidal activity. With $^1O_2$ previously being shown as a non-specific method to kill bacteria, this treatment method could be viable for a host of microorganisms.

3. Conclusion

This study described an approach to functionalizing SiO_2 with photocatalytic coatings targeted at antibacterial applications using visible light. Synthesis of poly(hydroxy ethyl acrylate) polymer brushes and crosslinking via carboxylic acid-functionalized porphyrins (TCPP) through esterification was described. Successful synthesis was verified by characterization with ellipsometry, X-ray photoelectron

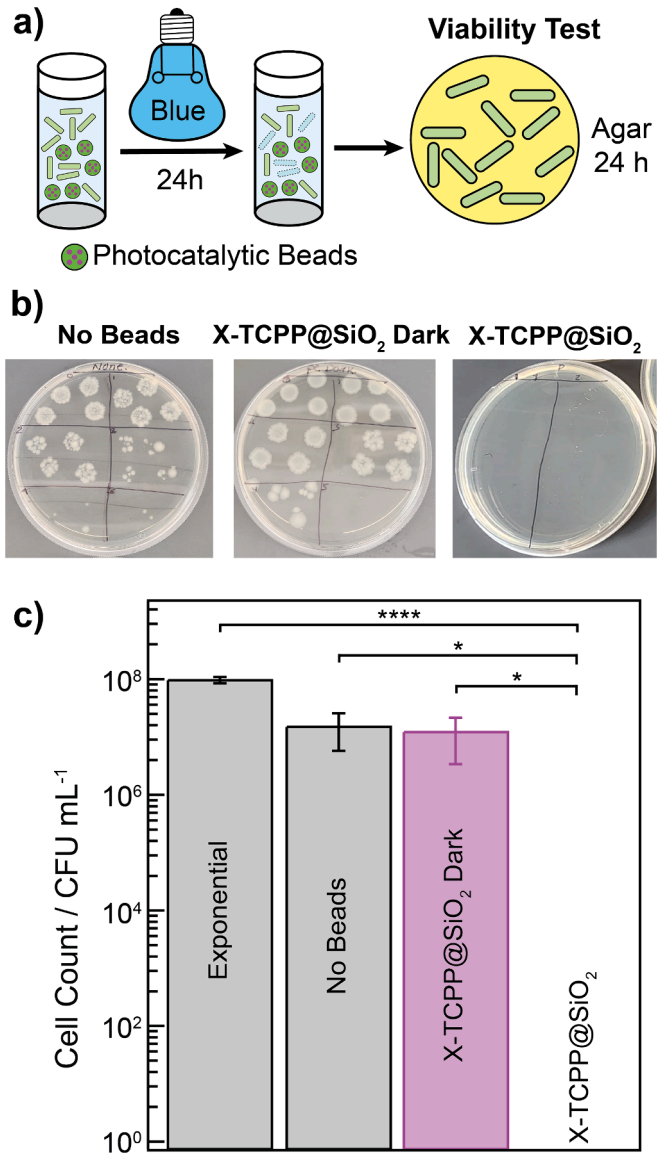

Fig. 5. (a) Scheme of viability tests conducted on photocatalytic SiO_2 beads. (b) Viability tests from no beads, X-TCPP@SiO_2 beads with no light, and X-TCPP@SiO_2 beads in the presence of blue light (left to right). (c) Cell count as determined from viability tests for different treatment conditions, $*p < 0.1$, $****p < 0.0001$.

spectroscopy, and ^{1}H NMR spectroscopy on flat and spherical SiO_2 surfaces. SiO_2 beads that were modified with TCPP-bearing polymer brush coatings showed production of singlet oxygen ($^1O_2$) upon exposure to visible (blue) light. $^1O_2$ production rates were higher than control experiments (e.g., non-functionalized SiO_2). Resulting from this capacity to generate $^1O_2$, the modified photoactive materials exhibited antibacterial activity against *E. coli* in both live/dead assay and cell viability tests.

CRediT authorship contribution statement

Brock Hunter: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Jessica L. Sacco:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Kira Katterle:** Investigation, Formal analysis. **Joy Kirigo:** Writing – review & editing, Methodology, Investigation, Data curation. **Thomas K. Wood:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Formal analysis. **Esther W. Gomez:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Christian W. Pester:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Esther W. Gomez reports financial support was provided by National Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper].

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113090.

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